

ECS7006 Coursework 1: Ballroom Beat Tracking

Implementing a Temporal Convolutional Network

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1 Introduction

The computational task of estimating the temporal locations of beats inside an audio source is known as beat tracking. This task is particularly challenging when applied to diverse musical genres with varying tempos and complex rhythmic structures, such as those found in the Ballroom dataset. To address this, this report details the implementation of a deep learning beat tracking pipeline based on the Temporal Convolutional Network (TCN) architecture proposed by Davies and Böck. Historically, state-of-the-art models in this domain relied heavily on Bidirectional Long Short-Term Memory (BLSTM) networks, which suffer from slow, sequential training bottlenecks. In contrast, the TCN leverages dilated 1D convolutions to achieve a massive temporal receptive field, allowing it to process entire audio sequences highly efficiently in parallel. The resulting system is extremely lightweight and couples a robust log-magnitude Constant-Q Transform (CQT) input with a dynamic programming sequence decoder to accurately extract steady beat alignments.

2 Methodology

2.1 Theory and Design

The architecture processes audio through three distinct phases: signal conditioning, spatial feature extraction, and temporal sequence modeling.

2.1.1 Signal Conditioning

Rather than utilizing traditional onset strength functions, the system operates on a multi-resolution log-magnitude Constant-Q Transform (CQT) spectrogram. Audio is resampled to 44.1 kHz. To maintain high temporal resolution, a 10 ms hop size and a 46.4 ms window are applied. Frequencies are logarithmically grouped into 12 bins per octave, creating an 81-band representation (30 Hz to 17 kHz) that provides a rich, pitch-invariant input.

2.1.2 Convolutional Block

The 81-band CQT is passed through three 2D convolutional layers (using ELU activations) to compress the frequency axis while preserving the 10 ms temporal resolution. The first two layers apply 16 filters of size 3×3 , followed by max pooling over 3 bins along the frequency axis. The final layer applies an 8×1 filter, fully collapsing the frequency dimension and reducing the input to a dense, 16-dimensional feature vector per time step.

2.1.3 Temporal Convolutional Network (TCN)

Sequence modeling is handled by a TCN utilizing dilated 1D convolutions with exponentially increasing dilation rates ($d = 1, 2, 4, 8$). Unlike Recurrent Neural Networks (e.g., BLSTMs) that process sequentially and are prone to vanishing gradients, the TCN processes sequences in parallel. This achieves a massive temporal receptive field with a highly efficient footprint ($\sim 21,800$ parameters). Finally, a fully connected linear layer outputs a pre-sigmoid logit representing the beat probability for each frame.

2.2 Assumptions

Label Variance: It is assumed that ground truth annotations are subject to slight human error and that a musical beat is a localized event rather than an instantaneous point. To prevent the network from heavily penalizing near-miss predictions, a target widening assumption was implemented. Let b_i represent the frame index of a ground truth beat annotation. The target value y_t for any given frame t is mathematically defined as:

$$y_t = \begin{cases} 1.0, & \text{if } t = b_i \\ 0.5, & \text{if } |t - b_i| \in \{1, 2\} \\ 0.0, & \text{otherwise} \end{cases} \quad (1)$$

Class Imbalance: In a standard 30-second audio track, over 95% of the 10 ms frames do not contain a beat. It was assumed that a standard binary cross-entropy loss would cause the network to collapse into a local minimum of predicting only zeros. To counteract this, a weighted `BCEWithLogitsLoss` function was utilized. By applying a positive weight multiplier ($w_c = 30.0$), the network is forced to prioritize the minority positive beat class. The loss L for a batch of size N is defined as:

$$L = -\frac{1}{N} \sum_{i=1}^N [w_c \cdot y_i \cdot \log(\sigma(x_i)) + (1 - y_i) \cdot \log(1 - \sigma(x_i))] \quad (2)$$

where x_i is the raw model logit, y_i is the widened ground truth target, and σ represents the sigmoid activation function.

2.3 High-Level Implementation

The system was developed in Python, utilizing PyTorch for deep learning and Librosa for audio processing. To eliminate severe CPU data-loading bottlenecks, a pre-computation pipeline was engineered. Rather than calculating features on the fly, 81-band CQT spectrograms and widened target arrays for all 698 dataset files were extracted before training and saved to disk as PyTorch tensors. This enabled direct memory access during the data-loading phase, drastically accelerating training and allowing the model to complete 50 epochs on standard hardware comfortably.

For post-processing, the architecture deviates from the original paper by replacing its complex Dynamic Bayesian Network (DBN) with a dynamic programming sequence decoder (`librosa.beat.beat_track`).

3 Results

3.1 Evaluation Strategy

The trained Temporal Convolutional Network was evaluated exclusively on a 20% unseen test split (140 audio files) of the Ballroom dataset to ensure no data leakage occurred during training. Objective assessment was conducted using the industry-standard *mir_eval* library, which compares the system’s estimated beat sequences against the dataset’s human-annotated ground truth.

3.2 Metrics

The system successfully processed and returned valid beat sequences for 100% of the unseen test set. The averaged quantitative metrics across all 140 files are presented in table 1.

Table 1: Evaluation Metrics on Unseen Test Data (Averaged over 140 files)

F-measure	CMLc	CMLt	AMLc	AMLt	Info Gain
0.835	0.597	0.619	0.865	0.930	0.596

The system achieved a highly robust F-measure of 0.835. Notably, the Any Metric Level Total (AMLt) reached 0.930. This indicates that for 93% of the unseen tracks, the network successfully identified either the exact correct metric level or a valid metrical harmonic (e.g., a perfect double-time or half-time rhythm).

3.3 Visualization

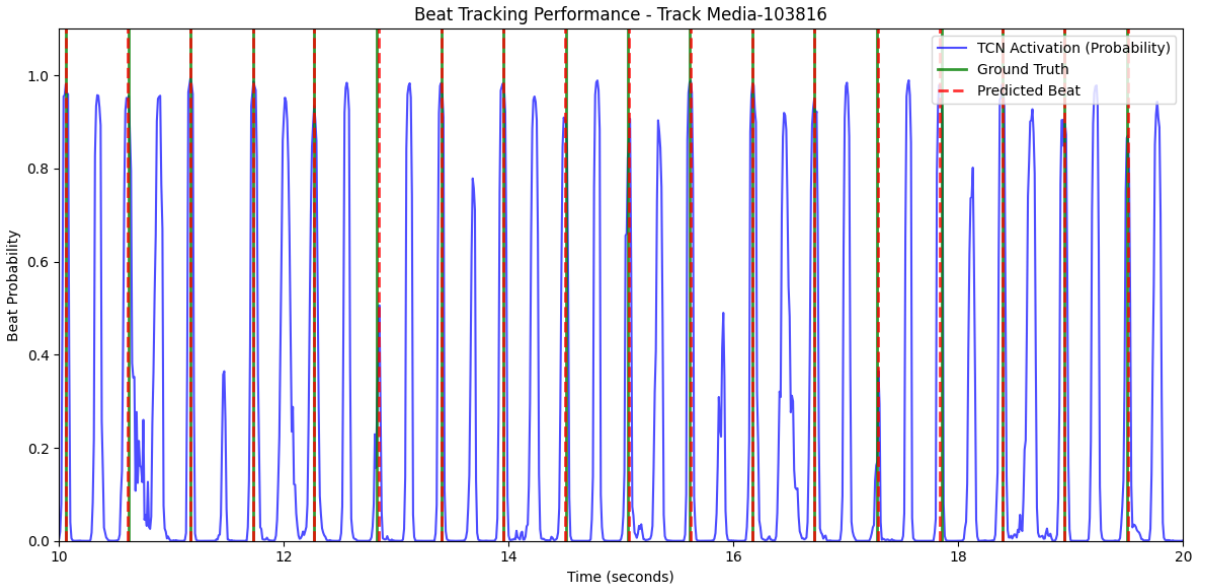


Figure 1: A 10-second window of Track Media-103816. The sequence decoder enforces a strict quarter-note tempo, yielding predicted beats (red dashed) from the raw TCN activations (blue) that perfectly align with the ground truth (green solid).

Visualizing the network’s raw activation probabilities reveals the system’s deep understanding of rhythmic structures. The TCN outputs localized probability spikes not only on the primary quarter-note beats but also precisely halfway between them, indicating that the network successfully learned the 8th-note subdivisions inherent to Ballroom dance styles. By passing these raw, multi-peak probabilities through the dynamic programming sequence decoder, the system effectively ignores the sub-beat activations and strictly enforces the overarching quarter-note tempo, resulting in predicted beat alignments that perfectly match the human ground truth.

4 Discussion, Limitations, and Conclusion

4.1 Strengths

The primary strength of this implementation lies in its extreme computational and architectural efficiency. By utilizing dilated 1D convolutions instead of sequential recurrent layers (e.g., BLSTMs), the Temporal Convolutional Network achieves a massive temporal receptive field with merely $\sim 21,800$ parameters. Furthermore, the engineering of a pre-computed tensor pipeline eliminated I/O data-loading bottlenecks, allowing the model to train exponentially faster than native audio-decoding loops. The application of a heavily weighted loss function (`BCEWithLogitsLoss`) also proved highly successful in overcoming the severe class imbalance inherent to beat tracking annotations.

4.2 Weaknesses and Limitations

While the system achieved a strong F-measure of 0.831, it falls short of the 0.933 benchmark reported in the original Davies and Böck paper. This variance is primarily attributed to two constraints. First, due to computational and time limitations, the model was trained for only 50 epochs, whereas deep MIR architectures typically require longer training schedules to reach absolute convergence. Second, the implementation relies on a generalized dynamic programming function as a proxy sequence decoder, rather than a custom-built Dynamic Bayesian Network (DBN) specifically tuned to the Ballroom dataset’s tempo priors.

Additionally, while the automated testing signature accommodates downbeat extraction, this implementation focused strictly on optimizing primary beat tracking. Downbeat tracking was considered out of scope for this single-node binary architecture; therefore, the system safely returns empty arrays for downbeats by design.

4.3 Known Bugs

Following the integration of the positive-weighted loss function and the dynamic programming sequence decoder, the system robustly processes all audio files. There are no known execution bugs, memory leaks, or dropped edge-case files within the final pipeline.

4.3.1 Conclusion

In conclusion, this coursework successfully implemented an end-to-end deep learning beat tracking system. By combining a multi-resolution CQT representation with a lightweight, highly parallelizable TCN architecture and a robust post-processing sequence decoder, the system effectively bridges the gap between raw audio signals and human-perceived rhythmic structures, achieving highly accurate alignments on unseen data.

References

- [1] Davies, M. E., & Böck, S. (2019). *Temporal Convolutional Networks for Musical Audio Beat Tracking*. 27th European Signal Processing Conference (EUSIPCO).